Problem 7.10

Suppose $a(t) = \alpha t^2 + 10\beta$. If \$X invested at time 0 accumulates to \$1,000 at time 10, and to \$2,000 at time 20, find the original amount of the investment X.

Solution

Since the same initial investment X accumulates to both values,

$$X\,a(10) = 1000 \qquad \text{and} \qquad X\,a(20) = 2000.$$

Therefore,

$$\frac{1000}{a(10)} = \frac{2000}{a(20)}.$$

Now,

$$a(10) = 100\alpha + 10\beta, \qquad a(20) = 400\alpha + 10\beta.$$

So,

$$\frac{1000}{100\alpha + 10\beta} = \frac{2000}{400\alpha + 10\beta}.$$

Cross-multiplying gives

$$2000(100\alpha + 10\beta) = 1000(400\alpha + 10\beta).$$

Dividing by 1000,

$$2(100\alpha + 10\beta) = 400\alpha + 10\beta.$$

Hence,

$$200\alpha + 20\beta = 400\alpha + 10\beta,$$

so

$$10\beta = 200\alpha, \qquad \text{that is,} \qquad \beta = 20\alpha.$$

Also, since an accumulation function satisfies $a(0) = 1$,

$$a(0) = 10\beta = 1.$$

Thus,

$$\beta = 0.1, \qquad \alpha = \frac{\beta}{20} = 0.005.$$

Now,

$$a(10) = 100(0.005) + 10(0.1) = 0.5 + 1 = 1.5.$$

Therefore,

$$X = \frac{1000}{1.5} = 666.67.$$

So the original amount of the investment was

$$\boxed{\$666.67}.$$